



Closed-Form Plug-In Bandwidth Selection for Multivariate Kernel Density Estimation

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Abstract

We derive a closed-form expression for the integrated squared Laplacian roughness of an isotropic multivariate Gaussian kernel density estimator, enabling non-iterative bandwidth selection in any dimension.

Keywords bandwidth selection, kernel density estimation, plug-in method, multivariate, nonparametric

1. Introduction

Kernel density estimation is fundamental to nonparametric statistics. Given observations $\mathbf{X}_1, \dots, \mathbf{X}_n \in \mathbb{R}^d$, the Gaussian kernel density estimator with bandwidth matrix $\mathbf{H}$ is $\hat{f}_{\mathbf{H}}(\mathbf{x}) = n^{-1} \sum_{i=1}^n K_{\mathbf{H}}(\mathbf{x} - \mathbf{X}_i)$. The quality of the estimator depends critically on $\mathbf{H}$.

Matrix bandwidth selectors (Duong and Hazelton, 2003; Chacón and Duong, 2010) require iterating over $d(d+1)/2$ parameters. Rules of thumb (Scott, 1992; Silverman,

1986) assume Gaussian reference densities and oversmooth multimodal data. In one dimension, the Sheather–Jones plug-in selector (Sheather and Jones, 1991) provides an effective middle ground by estimating the roughness of f'' from the data. The core of the method is a closed-form evaluation of the Gaussian roughness integral. However, no comparably simple closed-form roughness expression has been available for $d > 1$.

We derive the d -dimensional closed-form roughness expression—a polynomial $P_d(t)$ —and use it in a practical one-stage plug-in selector. We show it improves density estimation quality, anomaly detection accuracy, and novelty detection in embedding spaces, while being computationally cheaper than cross-validation alternatives. The method is released as the open-source `gsj` package.

2. Background

2.1. AMISE in whitened coordinates

We work in whitened coordinates: $\mathbf{Y}_i = \hat{\Sigma}^{-1/2} \mathbf{X}_i$ where $\hat{\Sigma}$ is the sample covariance. In this space the bandwidth is isotropic ($\mathbf{H} = h^2 \mathbf{I}_d$) and the asymptotic mean integrated squared error is

$$\text{AMISE}(h) = \frac{(4\pi)^{-d/2}}{nh^d} + \frac{h^4}{4} \Psi, \quad \Psi = \int [\nabla^2 f(\mathbf{x})]^2 d\mathbf{x}. \quad (1)$$

Minimizing gives $h^* = (d(4\pi)^{-d/2}/(n\Psi))^{1/(d+4)}$.

Remark 1 The Laplacian roughness functional Ψ is the correct bias term specifically because we work in whitened coordinates where $\Sigma = \mathbf{I}_d$. In original coordinates with $\mathbf{H} = h^2 \hat{\Sigma}$, the bias involves $\text{tr}\{\hat{\Sigma} \nabla^2 f\}$, which reduces to $\nabla^2 f$ after whitening.

2.2. Relationship to the Sheather–Jones algorithm

The full Sheather and Jones (1991) algorithm uses a two-stage pilot: estimate ψ_6 via normal reference, derive pilot g_2 , estimate ψ_4 using g_2 , then plug into the AMISE formula. This achieves convergence rate $O(n^{-4/13})$ (Hall, Sheather, Jones and Marron, 1991).

Our approach uses a one-stage Silverman pilot—the same construction implemented in R’s `locfit::sjpi` (Loader, 1999). This sacrifices the optimal rate (achieving $O(n^{-5/14})$ in one dimension) but gains simplicity and direct generalizability to d dimensions. The $(d + 4)$ -th root dampens pilot errors: a 20% error in $\hat{\Psi}$ produces only $\sim 3\text{--}4\%$ error in $\hat{h}$.

3. Derivation

All derivations assume whitened data and isotropic bandwidth h_0 .

Theorem 1 (Kernel Laplacian) *For the isotropic Gaussian kernel $K_h(\mathbf{t}) = (2\pi h^2)^{-d/2} \exp(-\|\mathbf{t}\|^2 / (2h^2))$, the Laplacian is $\nabla^2 K_h(\mathbf{t}) = h^{-2} K_h(\mathbf{t})(\|\mathbf{t}\|^2 / h^2 - d)$.*

Theorem 2 (Product of Gaussians) $K_h(\mathbf{x} - \mathbf{a})K_h(\mathbf{x} - \mathbf{b}) = C_{\mathbf{ab}} \cdot \phi_{h/\sqrt{2}}(\mathbf{x} - (\mathbf{a} + \mathbf{b})/2)$ where $C_{\mathbf{ab}} = (4\pi h^2)^{-d/2} \exp(-\|\mathbf{a} - \mathbf{b}\|^2 / (4h^2))$.

Theorem 3 (Main result) *Let $r_{ij}^2 = \|\mathbf{Y}_i - \mathbf{Y}_j\|^2 / h_0^2$. The pairwise roughness contribution is*

$$I_{ij} = (4\pi h_0^2)^{-d/2} \exp(-r_{ij}^2/4) \cdot P_d(r_{ij}^2), \quad (2)$$

where

$$P_d(t) = \frac{t^2}{16} - \frac{(d+2)t}{4} + \frac{d(d+2)}{4}. \quad (3)$$

Proof Define $\mathbf{U} = \mathbf{Z} - \mathbf{Y}_i$, $\mathbf{V} = \mathbf{U} - \boldsymbol{\delta}$ with $\mathbf{U} \sim \mathcal{N}(\boldsymbol{\delta}/2, (h_0^2/2)\mathbf{I}_d)$. Expand $\mathbb{E}[L(\mathbf{U})L(\mathbf{V})]$ using $\mathbb{E}[\|\mathbf{W}\|^4] = d(d+2)$ for $\mathbf{W} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. Full moment calculations are given in the Appendix. $\square$

Corollary 4 *When $d = 1$: $P_1(t) = t^2/16 - 3t/4 + 3/4$, recovering the one-dimensional Gaussian roughness integral used by `locfit:sjpi`. Verified numerically: relative difference $< 10^{-14}$.*

The complete roughness estimate and plug-in bandwidth are:

$$\hat{\Psi}(h_0) = \frac{1}{n^2(4\pi)^{d/2}h_0^{d+4}} \sum_{i,j} e^{-r_{ij}^2/4} P_d(r_{ij}^2), \quad \hat{h} = \left(\frac{d}{n\hat{\Psi}(4\pi)^{d/2}} \right)^{1/(d+4)}. \quad (4)$$

4. Computational methods

The exact computation requires $O(n^2d)$ operations. For larger datasets we provide four strategies: random subsampling ($O(md)$, independent of n); spatial truncation via KD-tree; FFT-based binning for $d \leq 3$; and GPU parallelization.

Since $|P_d(r^2)| \exp(-r^2/4)$ achieves its maximum at $r = 0$ and is monotonically decreasing for $r \geq 3$, truncating pairs beyond $8h_0$ introduces relative error below 2.4×10^{-5} for all $d \leq 10$.

5. Theoretical properties

Theorem 3 provides the exact value of $R(\nabla^2 \hat{f}_{h_0})$ for any finite sample—a deterministic identity, not an approximation.

The Silverman pilot $h_0 = O(n^{-1/(d+4)})$ gives $nh_0^{d+4} = 4/(d+2)$, a constant. Standard plug-in consistency arguments requiring $nh_0^{d+4} \rightarrow \infty$ do not directly apply. Formal

analysis at this critical rate is left for future work. Empirical evidence suggests stable, convergent behaviour as n increases across all tested distributions.

Based on analogy with the one-dimensional one-stage rate: $|\hat{h} - h^*|/h^* = O_p(n^{-5/(3d+14)})$ (conjecture).

6. Empirical evaluation

6.1. Anomaly detection

We fit kernel density estimators on normal-class training data and score test points by $-\log \hat{f}(x)$. On the MNIST Digits dataset ($d = 8$), our method achieves $\text{AUC} = 0.800$ versus 0.743 (Silverman) and 0.716 (Scott)—an absolute improvement of +5.7%.

6.2. Text embedding novelty detection

Using 20 Newsgroups documents embedded with MiniLM-L6-v2 ($d = 10$ via PCA), leave-one-category-out anomaly detection shows our method winning 16 of 20 comparisons (80%).

6.3. Comparison with cross-validation

Our method outperforms likelihood cross-validation by +0.9% AUC while being $2.7\times$ faster—notable given that we use a non-iterative closed-form selector.

7. Software

The `gsj` package is available at <https://pypi.org/project/gsj/> with source at <https://github.com/qzyu999/gsj>. It supports NumPy, PyTorch, and CuPy backends with automatic method selection based on dataset size.

8. Discussion

The method is most beneficial for multimodal data in $d = 2\text{--}15$. On unimodal Gaussian data it provides 5–15% worse integrated squared error than the optimal Silverman rule—an acceptable trade-off given gains of 65–90% on structured data.

Limitations include: scalar bandwidth only (anisotropic data may need diagonal selectors); curse of dimensionality for $d > 15\text{--}20$; and Gaussian kernel restriction.

Future work includes: two-stage pilot in d dimensions (requiring the multivariate ψ_6 functional); formal consistency proof via U-statistic theory; diagonal bandwidth extension; and application to density-based data preprocessing in machine learning pipelines.

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Data availability

All datasets used are publicly available via scikit-learn and the 20 Newsgroups corpus. Code and experiments are available at <https://github.com/qzyu999/sj>.

Appendix: Moment calculations

For $\mathbf{W} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$: $\mathbb{E}[\|\mathbf{W}\|^4] = d(d+2)$. For $\mathbf{U} = \boldsymbol{\mu} + \sigma\mathbf{W}$: $\mathbb{E}[\|\mathbf{U}\|^2] = \|\boldsymbol{\mu}\|^2 + d\sigma^2$; $\mathbb{E}[\|\mathbf{U}\|^4] = \|\boldsymbol{\mu}\|^4 + 2(d+2)\sigma^2\|\boldsymbol{\mu}\|^2 + d(d+2)\sigma^4$. With $\mathbf{V} = \mathbf{U} - \boldsymbol{\delta}$: $\mathbb{E}[\|\mathbf{U}\|^2\|\mathbf{V}\|^2] = \|\boldsymbol{\delta}\|^4/16 + (d-2)\|\boldsymbol{\delta}\|^2 h_0^2/4 + d(d+2)h_0^4/4$.

All verified symbolically (SymPy) and numerically (relative difference $< 10^{-14}$ between d -dimensional formula at $d=1$ and dedicated one-dimensional implementation; agreement with R's `locfit::sjpi` to within 0.03%).

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